Predictive Analytics in Genetic Disorders: Focusing on Hemophilia B

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Abstract

Accurate prediction of severity classes in Hemophilia B is critical for enhancing clinical decision-making and patient evaluation. Traditional methods primarily focus on analyzing the biological and structural characteristics of protein mutations. However, these approaches often encounter challenges due to the complex, high-dimensional nature of the data, leading to issues such as overfitting and the inability to capture key nonlinear interactions, ultimately affecting prediction accuracy. This study introduces a robust machine learning framework aimed at addressing these limitations. The proposed framework incorporates advanced data preprocessing techniques, systematic feature selection, and an ensemble-based Bi-LSTM predictive model. By effectively managing heterogeneous data types and optimizing the handling of complex interactions, the framework improves prediction reliability and accuracy. Moreover, the model achieves better generalization, enhanced computational efficiency, and faster processing times, making it highly suitable for analyzing intricate biological datasets and providing actionable insights into disease severity classification.

Keywords: Hemophilia B, Machine Learning, Random Forest, severity

1 Introduction

The spread of genetic disorders, both globally and in India, is a major healthcare challenge. In India alone, about 5–7% of the population is affected by rare genetic diseases, making it crucial to predict disease severity accurately for early diagnosis and effective treatment. Early identification and intervention can drastically improve patient outcomes and reduce the strain on healthcare systems. This analysis focuses on a range of critical biological and structural parameters that influence disease progression. By leveraging these complex features, the research aims to improve the accuracy of severity predictions, contributing to more personalized and effective treatment approaches. This work is essential for advancing our understanding of genetic disorders and optimizing patient care.

Various machine learning and statistical techniques have been employed in current approaches to predict the severity of hemophilia. For classification tasks involving clinical and genetic data, traditional methods like “Support Vector Machines” (SVMs), “Random Forest”, and “Logistic Regression” are popular. More recent advancements incorporate deep learning approaches like “Recurrent Neural

Networks" (RNNs) and "Long Short-Term Memory" (LSTM) networks. Studies such as the one by Zhang et al.[1] have also explored deep learning frameworks like CNNs combined with network-based information to improve the prediction accuracy for genetic disorders. Additionally, Bi-LSTM has been applied to model DNA sequences and predict severity in diseases like prostate cancer, demonstrating its potential for sequence-based data analysis, as seen in Abass et al.'s [2] work.

In the domain of gene sequence analysis for hemophilia, these models attempt to capture complex patterns in genetic mutations and their impacts on protein function. Other approaches involve combining sequence- and network-based information to better understand the impact of mutations, as proposed in related studies like Shahid et al.'s [3] predictions using LSTM and Bi-LSTM for other genetic disorders.

High-dimensional datasets with both continuous and categorical variables, such as the protein sequence data used in this investigation, are best suited for Random Forest. Random Forest improves performance by building several decision trees and combining their forecasts. In order to capture intricate, non-linear correlations between input features and the target variable, Random Forest assesses the significance of the dataset's features, which include structural traits, hydrophobicity, and conservation scores. With a 75% prediction accuracy for the severity of genetic defects based on protein-related variables, the model is able to generalize well due to its capacity to avoid overfitting.

The next section contains a literature review of models used in genetic disease severity prediction, followed by an exploration of how Random Forest was chosen for this task. We discuss the dataset cleaning process, feature engineering, handling categorical variables, and scaling. The experimental setup section outlines the dataset, chosen hyperparameters, and computational environment. Results are presented, and insights drawn from model performance. Finally, the conclusion highlights the Random Forest model's effectiveness and suggests directions for future research.

2 Literature survey

2.1 SVM

During Your Activities Have some of you ever come across the term "Support Vector Machine (SVM)?" SVM has been used successfully in math for classifying and predicting (among others) medical conditions on various datasets [4-6]. Ferreira-Martins et al. [7] used network centrality, structural characteristics, and evolutionary information as input features for SVMs to predict FV-deficiency. The use of machine learning models directly implemented and trained in three areas, namely network centrality, structural features, and evolutionary data was also discussed. SVM showed AUC results of more than 0.65. David et al. [8] reported over 87% accuracy using SVM to classify 3000 FISH signals. SVMs, based on structural risk minimization, balance model performance and complexity, reducing overfitting in

diverse medical datasets. Ghazal et al. [9] claimed that the performance of SVM surpassed KNN in the prediction of cancer, dementia, and diabetes (respectively) with 92.8% accuracy.

2.2 XG Boost

Li et al. [10] in 2019 had the XGBoost algorithm predicting gene expression values from RNA-Seq and GEO datasets. The outcome was the lowest errors in both the validation (0.280) and test (0.282) sets among other models tested in the research. The class of research work from 2023 [11] that was on the line of cancer types classification on the genetic lesions basis reported the XGBoost performance out of the others. It got a balanced accuracy (BACC) score of 77% and an AUC of 97%. The BACC scores were elevated to 86% by the classification of 18 cancer types into groups of which the AUC was over 96%. Thus, the results indicate that the method could be used to enhance cancer diagnosis. XGBoost utilizes a boosting technique that builds multiple decision trees sequentially. Errors made by previous trees are corrected by every tree which improves accuracy and robustness. For complicated datasets like gene-disease relationships, this iterative method is especially helpful in improving predictions and minimizing overfitting. Kumar et al. [12] focused on the classification task of gene-disease associations, based on their semantic types. The XGBoost Classifier achieved an accuracy of 0.93, which is the best-performing model in the study.

2.3 Bi-LSTM

Several studies [13-16] have utilized models like BiLSTM for time-series forecasting and classification tasks in different medical contexts. In one study by Aung et al. [17] focused on forecasting daily COVID-19 cases that were confirmed, BiLSTM models outperformed the traditional ARIMA model, specifically in terms of error reduction, with Model 1 showing more reliable estimates across various infection stages and Model 2 excelling in countries with higher daily cases. For detecting genetic disorders using genomic data, a certain study by Nandhini et al. [18] made use of the Stacked ResNet-BiLSTM model and achieved a high accuracy of 99.85% along with strong precision and AUC scores. A third study [19] focused on EEG signal classification for MCI detection, where BiLSTM with multitaper spectral analysis achieved the highest performance with 98.97% accuracy, outperforming other machine learning models like decision tree and SVM.

2.4 Random Forest

Random Forest model has demonstrated the effectiveness in classifying medical datasets for many years. Alam et al. [20] evaluated the RF on heart disease, breast cancer and diabetes datasets, where it achieved accuracies of 84.5%, 96.2%, and 77.8% respectively where feature ranking was also applied to improve accuracy and reduce complexity. Another study by Stafford [21] focused on predicting disease phenotypes and treatment responses, such as Crohn’s disease vs. ulcerative colitis, where RF achieved accuracies between 80% and 90%, especially in high-dimensional

data scenarios. Lopes et al. [22] proposed an ensemble model for classifying the severity of HB cases, HemB-Class (RF and XGBoost), which reached an accuracy of over 80% with 0.71 sensitivity and 0.89 specificity. Tasin et al. [23] presented a study where the RF model outperformed SVM and k-NN in detecting cataracts from clinical and imaging data, achieving an accuracy of 94.2% due to its ability to handle complex, high-dimensional data. The proposed model may struggle with imbalanced data since it could have difficulty predicting the minority class. With high-dimensional sparse data, the model may overfit or miss important patterns. For time series data, it lacks the ability to handle temporal dependencies or sequential patterns effectively.

3 Dataset

The study [19] that was published in Frontiers in Bioinformatics (2022) provided the dataset for this project. It contains mutation data related to Hemophilia B, where the goal is to predict the severity of the condition based on point mutations. The dataset included various features capturing protein function scores, mutation impact predictors like SIFT, PROVEAN, and PolyPhen-2 and other biologic attributes such as amino acid changes and protein structure metrics. These features are the basis for predicting disease severity, helping to make clinical decisions about genetic disorders. The dataset appeared as a single-column structure, when initially loaded, due to parsing issues. This issue was resolved by splitting the column using a tab delimiter (\t), resulting in 33 distinct columns and 392 rows.

In order to ascertain if a mutation is detrimental or neutral, PROVEAN (Protein Variation Effect Analyzer) forecasts how amino acid substitutions or indels (insertions/deletions) would affect protein function. PolyPhen-2 (Polymorphism Phenotyping v2) uses sequence alignment and structural features to predict mutation damage and assesses the potential effects of amino acid substitutions on protein structure and function. Damage is more likely when scores are higher. By examining conserved areas, SIFT (Sorting Intolerant From Tolerant) predicts the functional impact of amino acid alterations and uses thresholds to categorize mutations as neutral or detrimental.

AreaSAS (Solvent Accessible Surface Area) is the surface area of proteins accessible to solvents which reveals the presence of newly exposed residues in the protein surface that can interact with other molecules. relSESA (Relative Solvent-Excluded Surface Area) is an index of how buried are residues inside the protein's core, and high relSESA values often contribute to stability. AreaSES (Solvent-Excluded Surface Area) is obtained by eliminating the cocrystal solvent, i.e., measurement of the excluded area. It gives a more precise packing study and can be used to understand how mutations affect protein interactions and epitope formation.

ConSurfDB [24] offers evolutionary conservation scores for amino acid residues in polypeptides and allows one to establish which of them are retained across various species. Such conservation scores reflect the probability that a protein residue is evolutionarily optimized to perform a pivotal function or maintain the protein's

structural configuration. ConSurfDB helps the users in locating specific important sites that potentially influence protein interactions or activity and help in investigating disease-related mutations.

Burts, evaluates the extent of one residue being covered by the architecture of the protein in the coat of three dimensional shape spans. It is in this context that this feature estimates how much residue of an amino acid is enveloped by other amino acids in contrast with being open to the solvent environment. Often but not always buried residues are found within the folds of the protein and more often than not, are associated with its stability or resistance to folds. Understanding buriedness bonds help the scientist to know the amino acid residues that are most likely not to be influenced by external factors but maintain the integrity of the structure.

In a three-dimensional protein model, KCore is a structural characteristic that counts the number of edges or bonds that are attached to a residue. It calculates the connectivity of the residue's k-core; greater k-core values signify saturation with k residues. Protein folding and function depend on these residues, and mutations in these areas can cause structural network disruption and functional impairment.

The maximum probability score from PolyPhen-2, or PolyPhen2-idPmax, indicates the possibility that a mutation will be harmful based on sequence and structure. The lowest quality score, PolyPhen2-idQmin, evaluates the degree of confidence in benign predictions. Reduced confidence is indicated by a low idQmin score, which implies that a mutation may still affect protein function. Prioritizing mutations for additional functional investigation is made easier by these characteristics.

4 Preprocessing Steps

In this section, the dataset was cleaned, standardized, and transformed for machine learning. Missing numeric data was imputed using the mean strategy, while categorical variables like AA_HGVS, AA_Legacy, and Domain underwent one-hot encoding. Numeric columns were standardized to prevent biases from varying scales. Feature selection was performed using the SelectKBest method with an ANOVA F-test to evaluate each feature's relationship with the target variable. The top 20 features with the highest F-scores (Fig. 1) were selected, significantly contributing to predicting Reported_Severity, ensuring the dataset focused on features most relevant to the classification task.

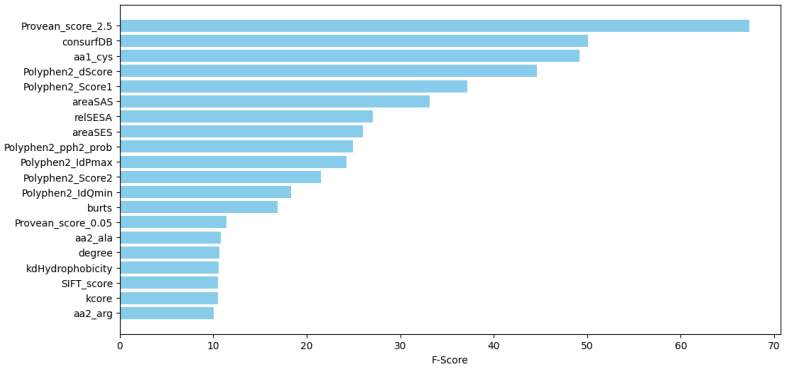


Fig. 1. Top 20 features based on their F-scores

Because Random Forest works well with high-dimensional data and can handle both continuous and categorical variables, we chose it for this dataset. By averaging forecasts from several decision trees, its ensemble nature improves accuracy by preventing overfitting. Feature extraction and model tuning were done in a methodical manner, which included standardizing features, converting category features, and imputing missing values. The dataset was made simpler by feature selection methods and dimensionality reduction. To improve Random Forest's accuracy and performance, hyperparameter optimization was carried out, guaranteeing that the model successfully fits the data for precise predictions.

4 Result and Analysis

The results indicate that the “Random Forest” model achieved the highest accuracy of 0.75, outperforming the other models in this comparison. The “precision”, “recall”, and F1 scores for both the "others" and "severe" classes are relatively balanced, with values of 0.76 and 0.73, respectively, for both precision and recall. This balance highlights the model’s robustness in distinguishing between different severity levels of hemophilia, making it a strong candidate for accurate predictions. Additionally, Random Forest’s ability to handle complex, high-dimensional data likely contributed to its superior performance.

Comparatively, the SVM model had an accuracy of 0.73 but showed a larger discrepancy in recall between the "others" (0.81) and "severe" (0.64) classes, indicating that it was better at identifying less severe cases but struggled with more severe cases. XGBoost performed slightly better than SVM with an accuracy of 0.74, but like SVM, it had a noticeable imbalance in recall between the "others" (0.78) and "severe" (0.69) classes. Bi-LSTM, while slightly less accurate at 0.659, had consistent precision and recall across both classes, though the overall scores were lower compared to the other models. This suggests that while Bi-LSTM maintains balance, it lacks the predictive strength seen in Random Forest and XGBoost. For the Random

Forest model, hyperparameter tuning was conducted using GridSearchCV [25]. The final selected parameters are elaborated in Table 1.

Table 1: Optimal Hyper parameters for Random Forest

Parameter	Value
bootstrap	True
max_depth	10
max_features	sqrt
min_samples_leaf	2
min_samples_split	2
n_estimators	100

This configuration resulted in a test accuracy of 0.75 and an execution time of 5 minutes 12 seconds. In comparison, the XGBoost model achieved a similar accuracy of 0.75 but took longer to execute, with a runtime of 8 minutes 33 seconds. The optimal hyperparameters for XGBoost were set as detailed in Table 2.

Table 2: Optimal Hyper parameters for XG Boost

Parameter	Value
colsample_bytree	0.6
gamma	0.5
learning_rate	0.01
max_depth	3
subsamples	0.8
n_estimators	200

While both models achieved the same accuracy, Random Forest was computationally more efficient for this dataset, making it the preferred choice in terms of speed without sacrificing performance (see Fig. 2).

The experiments were conducted on a machine with an Intel Core i7 (8 cores), 32GB of RAM, and running Ubuntu 20.04. Different system configurations

may lead to variations in execution times and performance due to differences in hardware and software environments.

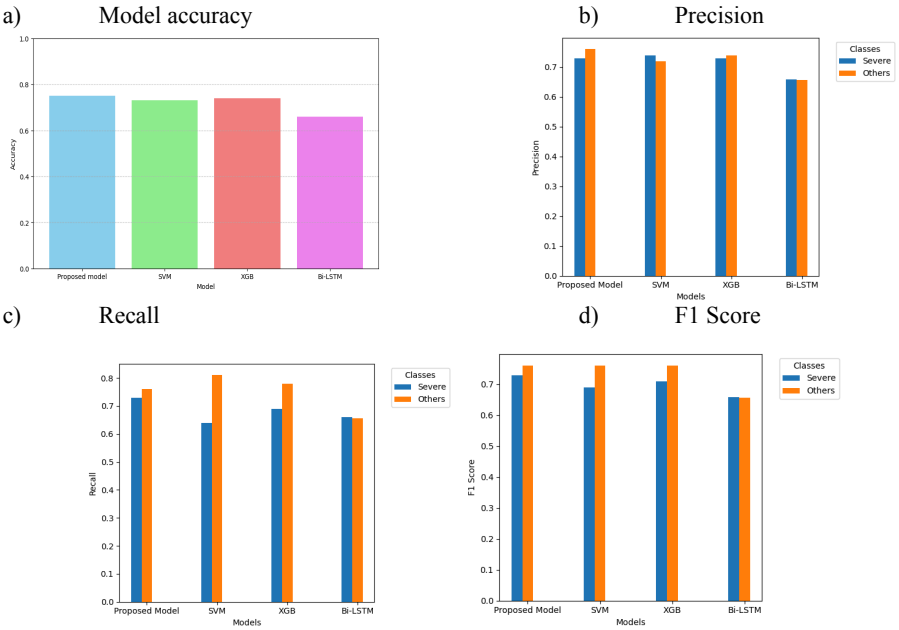
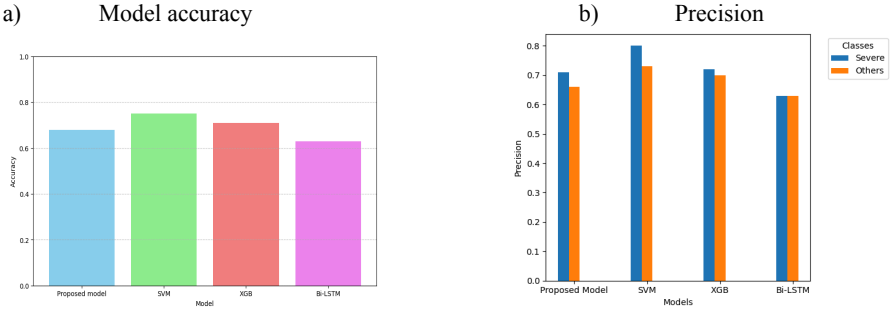
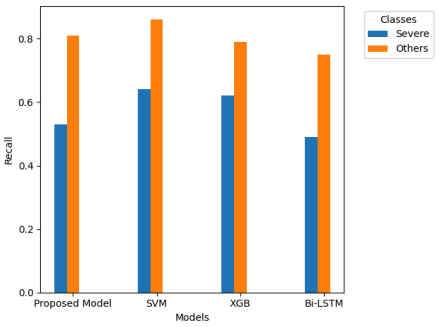


Fig. 2. Classification report for data with preprocessing (a) Model Accuracy, (b) Precision, (c) Recall, (d) F1 Score across different models (Proposed Model Random Forest, SVM, XGBoost, Bi-LSTM)

The analysis of classification metrics (Fig. 2 and Fig. 3) reveals notable insights into model performance both with and without preprocessing. The proposed model consistently outperforms SVM, XGBoost, and Bi-LSTM across all metrics when preprocessing is applied, except for recall. This is likely because the model is more conservative in identifying false negatives, especially in the "Others" class. It excels in precision, particularly in identifying "Severe" cases, highlighting its ability to minimize false positives. Without preprocessing, SVM performs best across all metrics, with the highest recall and F1 scores for "Others" instances, indicating its superior ability to detect this class without extensive data cleaning. Interestingly, all models show higher recall and F1 scores for the "Others" class and higher precision for the "Severe" class, suggesting that models generally struggle more with detecting "Severe" instances. Bi-LSTM, in particular, performs the worst across both scenarios, indicating its limitations in handling the complexities of this dataset.



c) Recall



d) F1 Score

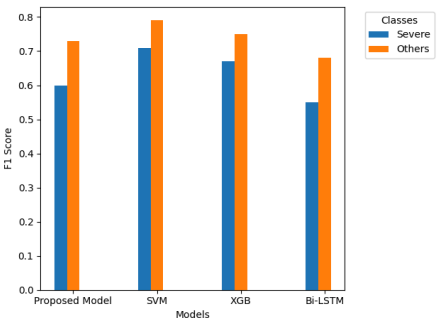


Fig. 3. Classification report for data without preprocessing (a) Model Accuracy, (b) Precision, (c) Recall, (d) F1 Score across different models (Proposed Model Random Forest, SVM, XGBoost, Bi-LSTM)

4. 1 Friedman Test

A Friedman Test [25] was used to evaluate model performance differences using cross-validation accuracies. Table 4 lists the parameters used in this analysis. 0.05 and 0.01 were selected as significance criteria because the dataset is medical in nature. For the purpose of identifying significant variations across repeated assessments, such as the cross-validation accuracies shown in Table 3, the Friedman test statistic (Q) quantifies rank variance between groups. By averaging across folds, cross-validation reduces the danger of overfitting and provides a reliable performance estimate. The total ranks allotted to each algorithm across datasets are shown by rank sums. Whether observed performance differences are statistically significant can be ascertained with the use of this statistical method.

Table 3: Cross Validation accuracies for 5 folds and Rank Sums

Fold	RF	SVM	XG	Bi-LSTM	RF	SVM	XG	Bi-LSTM
1	0.75	0.6709	0.6835	0.68	1	4	2	3
2	0.6667	0.7215	0.7342	0.63	3	2	1	4
3	0.65	0.5949	0.6456	0.68	2	4	3	1
4	0.6	0.6282	0.6795	0.72	4	3	2	1
5	0.6333	0.6538	0.6282	0.67	3	2	4	1

We observe that the test statistic (1839.0) is far greater than both critical values and hence we reject the null hypothesis at both significance levels. This means there is a significant difference in the performance ranks of the models.

Table 4: Parameter configuration for Friedman test

Models	4
Folds	5
Degrees of Freedom	3
Model 1	13
Model 2	15
Model 3	12
Model 4	10
Critical Value	11.34, 7.81
Friedman test statistic Q	1839.0
Significance Level	0.01, 0.05

5 Conclusion

In conclusion, this machine learning framework developed in the study's could be used to predict the severity of Hemophilia B using mutation data, with emphasis that using Random Forest produced the best results out of the models used. This is because the model is designed to contain high dimensional and a plethora of biological features such as the structural and functional properties of the proteins and other features which aid in the prediction of the disease severity. Random forest performed well, especially in diagnosing severe patients, where it was most critical, and maintained a fair precision recall tradeoff compared to the other models with hardly any superiority in terms of both speed and precision. The key inferences for this work are as follows: Random Forest model achieved the highest accuracy of 0.75 with lowest execution time of 312 seconds, with balanced precision and recall scores for both ‘others’ and ‘severe’ Its ability to handle complex, high-dimensional data resulted in faster and more accurate predictions compared to other models.

This work demonstrates how machine learning could be usefully employed in the clinics and aids in predicting the severity of a genetic disorder. This work may be extended into the future in response by investigating more sophisticated models or more biological information, in order to make better predictions. There is also a possibility that the expansion of the dataset and the use of deep learning may give new understanding to the links between genetic mutations and the disease that could help in developing more individualized solutions for the patients.

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